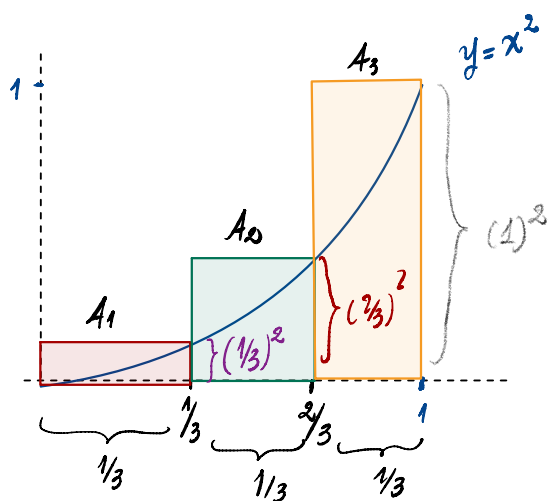




$$A \approx A_1 + A_2 + A_3$$

$$A_1 = \left(\frac{1}{3}\right)\left(\frac{1}{3}\right)^2$$

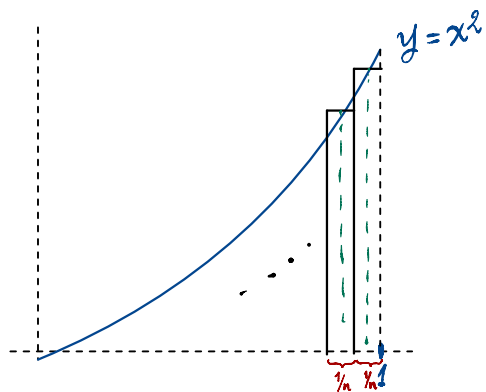
$$A_2 = \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^2$$

$$A_3 = \left(\frac{1}{3}\right)(1)^2$$

$$A \approx \frac{1}{3^3} + \frac{4}{3^3} + \frac{1}{3}$$

Using geometry we can check that  $A = \frac{1}{3}$ .

Example. Show that the area in the previous example is  $\frac{1}{3}$ .



$$A_n = \frac{1}{n} \left(\frac{1}{n}\right)^2 + \frac{1}{n} \left(\frac{2}{n}\right)^2 + \dots + \frac{1}{n} \left(\frac{n}{n}\right)^2$$

$$= \frac{1}{n} \left\{ \frac{1}{n^2} + \frac{2^2}{n^2} + \dots + \frac{n^2}{n^2} \right\}$$

$$= \frac{1}{n^3} \left\{ 1 + 2^2 + 3^2 + \dots + n^2 \right\}$$

$$= \frac{1}{n^3} \cdot \sum_{i=1}^n i^2$$

Since  $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$

So,

$$A_n = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6}$$

thus,

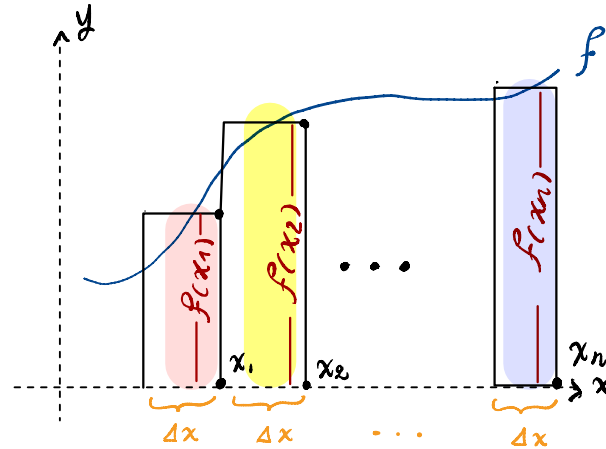
$$A = \lim_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} \frac{1}{n^3} \frac{n(n+1)(2n+1)}{6}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^3} \cdot \cancel{n} \cdot \cancel{n} (1 + 1/n) \cdot \cancel{n} (2 + 1/n)}{6}$$

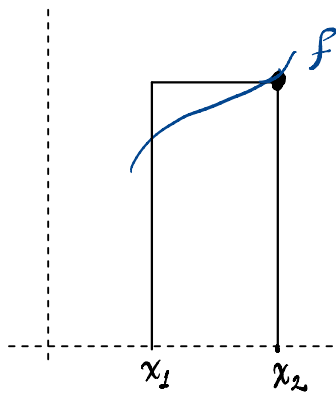
$$= \lim_{n \rightarrow \infty} \frac{(1 + 1/n)(2 + 1/n)}{6} = \frac{2}{6} = \frac{1}{3}$$

Def. The area  $A$  of the region that lies under the curve of a continuous function  $f$  is the limit of the areas of approximating rectangles.

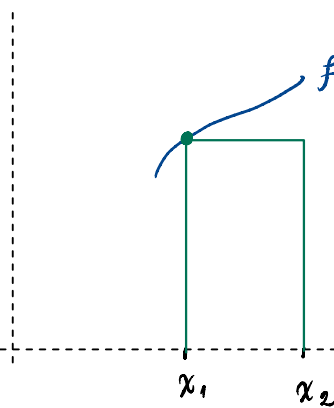
$$A = \lim_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} \{f(x_1)\Delta x + f(x_2)\Delta x + \dots + f(x_n)\Delta x\}.$$



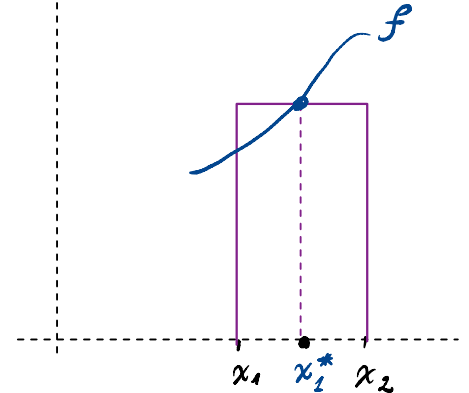
Upper-Sum.



Upper-Sum



low-sum



Riemann-Sum

In the Riemann sense we have:

$$A = \lim_{n \rightarrow \infty} \{f(x_1^*)\Delta x + f(x_2^*)\Delta x + \dots + f(x_n^*)\Delta x\}.$$

Example. Let  $A$  be the area of the region that lies under the graph of  $f(x) = \cos x$  between  $x=0$  and  $x=b$  where  $0 \leq b \leq \frac{\pi}{2}$ .

a). Using the right end points find the expression for  $A$  as a limit.

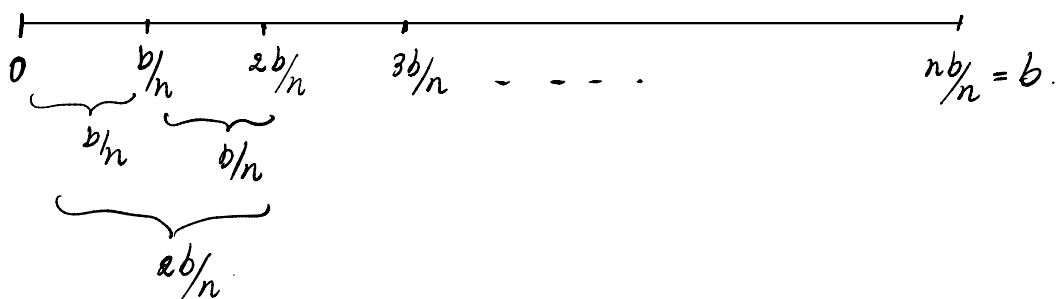
Do not evaluate that limit.

b). Estimate the area for  $b = \frac{\pi}{2}$  by taking the sample points to be midpoints.

Sln: a). Since  $a=0$  the width of the rectangles are:

$$\Delta x = \frac{b-a}{n} = \frac{b}{n}.$$

So,  $x_1 = \frac{b}{n}, x_2 = \frac{2b}{n}, x_3 = \frac{3b}{n} \dots x_n = \frac{nb}{n}.$



$$A_n = f(x_1)\Delta x + f(x_2)\Delta x + \dots + f(x_n)\Delta x$$

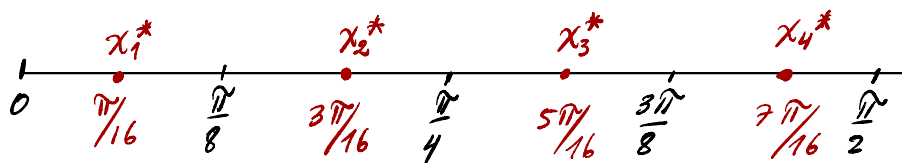
$$= \cos x_1 \cdot \Delta x + \cos x_2 \Delta x + \dots + \cos x_n \Delta x.$$

$$= \frac{b}{n} \left\{ \cos\left(\frac{b}{n}\right) + \cos\left(\frac{2b}{n}\right) + \cos\left(\frac{3b}{n}\right) + \dots + \cos(b) \right\}.$$

$$= \frac{b}{n} \sum_{i=1}^n \cos\left(\frac{bi}{n}\right)$$

$$A = \lim_{n \rightarrow \infty} \frac{b}{n} \sum_{i=1}^n \cos\left(\frac{bi}{n}\right).$$

b). with  $n=4$  and  $b = \frac{\pi}{2}$ .  $\Delta x = \frac{\frac{\pi}{2} - 0}{4} = \frac{\pi}{8}$  then.



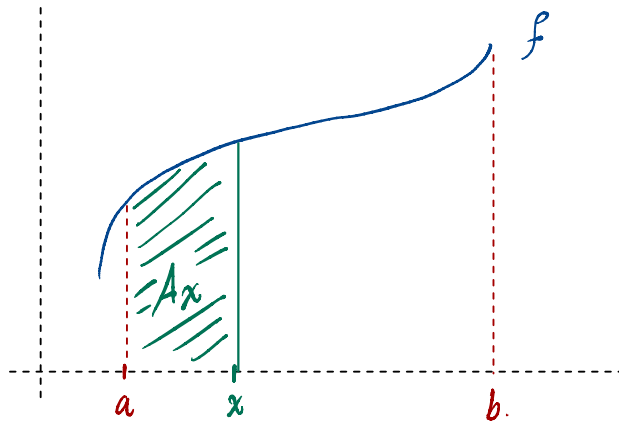
$$R_4 = \sum_{i=1}^4 f(x_i^*) \Delta x = \Delta x \cdot \sum_{i=1}^4 f(x_i^*)$$

$$= \frac{\pi}{8} \left( \cos\left(\frac{\pi}{16}\right) + \cos\left(\frac{3\pi}{16}\right) + \cos\left(\frac{5\pi}{16}\right) + \cos\left(\frac{7\pi}{16}\right) \right) \approx 1.006.$$

The Fundamental Theorem of Calculus.

\* The Fundamental theorem of Calculus part 1.

The first part of the FT deals with the function defined by an equation.



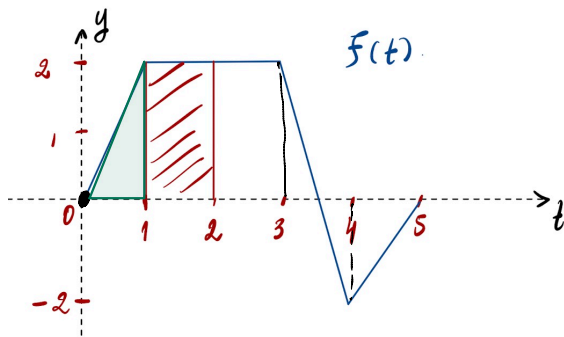
then, notice that this area depends on  $x$ .

$$g(x) = \int_a^x f(t) dt \longrightarrow \text{The integral of } f(t) \text{ between } a \text{ and } x.$$

$\underbrace{\hspace{10em}}$   
 area under the  
 curve  $f(t)$  between  
 $a$  and  $x$ .

Where  $f$  is a continuous function on  $[a, b]$  and  $a \leq x \leq b$ . Observe that  $g$  depends only on  $x$ . Which appears as the variable upper limit in the integral.

Example.



$$\text{If } g(x) = \int_0^x f(t) dt.$$

Find the values  $g(0)$ ,  
 $g(1)$ ,  $g(2)$ ,  $g(3)$ ,  $g(4)$   
 $g(5)$ .

Sln.

$$g(0) = \int_0^0 f(t) dt = 0$$

$$g(1) = \int_0^1 f(t) dt = \frac{1}{2} (1)(2) = 1.$$

$$g(2) = \int_0^2 f(t) dt = \int_0^1 f(t) dt + \int_1^2 f(t) dt = 1 + (1)(2) = 1 + 2 = 3.$$

$\underbrace{\hspace{10em}}$   


$\underbrace{\hspace{10em}}$   


$$g(3) = 3 + 2 = 5$$

$$g(4) = ?$$

$$g(5) = 5.$$



## The Fundamental Theorem of Calculus part 1.

If  $f$  is a continuous function on  $[a, b]$  then the function  $g$  defined by

$$g(x) = \int_a^x f(t) dt \quad ; \quad a \leq x \leq b$$

is continuous on  $[a, b]$  and differentiable on  $(a, b)$  and

$$g'(x) = f(x).$$

Example. Find the derivative of  $g(x) = \int_0^x \sqrt{1+t^2} dt$ .

Sln. Since  $f(t) = \sqrt{1+t^2}$  so

$$g(x) = \int_0^x f(t) dt.$$

So by FTC P1  $g'(x) = f(x) = \sqrt{1+x^2}$ .

Example. Find  $\frac{d}{dx} \int_1^{x^4} \sec t dt$ .

Sln: let  $f(t) = \sec t$ . but

$$g(x) = \int_1^{x^4} \sec t dt = h(s(x))$$

where  $h(x) = \int_1^x f(t) dt$  and  $s(x) = x^4$ .

$$h(s(x)) = h(x^4) = \int_1^{x^4} f(t) dt = g(x) \quad \checkmark$$

$$\underset{\text{C'R}}{g'(x)} = h'(s(x)) \cdot s'(x) = f(s(x)) \cdot s'(x) = 4x^3 \sec(x^4)$$

$$g(x) = \int_1^{x^4} \sec t dt \Rightarrow g'(x) = \sec(x^4) \cdot \frac{d}{dx}(x^4) = 4x^3 \sec(x^4).$$

Problem for you! Compute  $\frac{d}{dx} \int_{\sqrt{x}-x}^{x^2+4x+1} \cos(t) dt$ .

Properties: i).  $\int_a^a f(x) dx = 0$

ii).  $\int_a^b f(x) dx = - \int_b^a f(x) dx$ .

## The Fundamental Theorem of Calculus Part 2.

If  $f$  is a continuous function on  $[a, b]$  then

$$\int_a^b f(x) dx = F(b) - F(a).$$

where  $F$  is any antiderivative of  $f$ , i.e.,  $F$  is s.t.  $F' = f$ .

Example. Evaluate the integral  $\int_{-2}^1 x^3 dx$ .

Sln. We need an antiderivative for  $f(x) = x^3$ . ;  $F_1(x) = \frac{x^4}{4}$ .

$$\int_{-2}^1 x^3 dx = F_1(1) - F_1(-2) = \frac{1^4}{4} - \frac{(-2)^4}{4} = \frac{1}{4} - 4 = -\frac{15}{4}.$$

$$F_2 = \frac{x^4}{4} + \pi.$$

$$\begin{aligned} \int_{-2}^1 x^3 dx &= F_2(1) - F_2(-2) = \left( \frac{1^4}{4} + \pi \right) - \left( \frac{(-2)^4}{4} + \pi \right) \\ &= \frac{1^4}{4} + \cancel{\pi} - \frac{(-2)^4}{4} - \cancel{\pi} \\ &= -\frac{15}{4}. \end{aligned}$$

Example. Find the area under the parabola  $y = x^2$  for  $0 \leq x \leq 1$ .

Sln.  $A = \int_0^1 x^2 dx$  so,  $f(x) = x^2$   $F(x) = \frac{x^3}{3}$ . by FTC P2.

$$A = F(1) - F(0) = \frac{1}{3} - 0 = \frac{1}{3}.$$

Example: Find the area under the curve of cosine curve from 0 to  $b$  where  $0 \leq b \leq \frac{\pi}{2}$ .

Sln.  $A = \int_0^b \cos x dx = \sin x \Big|_0^b = \sin b - \sin 0 = \underline{\sin b}$

In particular if  $b = \frac{\pi}{2}$  then

$$A = \int_0^{\pi/2} \cos x dx = \sin \frac{\pi}{2} = 1.$$

Problem: What is wrong with the following calculation?

$$\int_{-1}^3 \frac{1}{x^2} dx = \frac{x^{-1}}{-1} \Big|_{-1}^3 = -\frac{1}{3} - 1 = -\frac{4}{3}.$$